

Education Cloud — V2 Weekly Execution PRD

Detailed dated week-by-week proposal for Phases 0–4 of *HOME_STUDY_COMPANION_V2_PRODUCT_PLAN.md* · Baseline 25 Aug 2026 · Status: Draft for review

1. Executive Summary

Problem. Grade-3 learners in connectivity-poor Kenyan households need a fun, honest, offline-first study companion; the V2 pivot ("Habit Shell Over Proven Core", D19) commits to evolving the existing Compose app rather than restarting.

Proposal. This PRD decomposes the phase-level plan (§9 of the parent document) into twelve dated weeks from the 25 Aug 2026 baseline, folding in work already landed and naming one owner-grade deliverable set plus exit criteria per week. It exists because no task-level weekly breakdown existed anywhere in the repo.

Already delivered before this baseline (not re-planned)

- Phase 1 core — HabitEngine (auto-freeze D10, mastery-only XP D11, milestones), Room v6 schema, milestone ceremony in quiz results, honest Progress/Profile screens — committed 25 Aug 2026 (4960c9f, 88efad0, 3ad2028); suite 76/76 green.
- Phase 2 core — /tutor/reexplain end-to-end with automatic validation replacing teacher review (cbab7d4, 26804cd).
- Plan §8 amended 25 Aug 2026: streak grace ships as one auto-freeze token per week; multi-day offline-awareness deferred (d92271f).

Business impact. (1) Fun loop usable offline end-to-end within 3 weeks — retention spine per D10–D13. (2) Trust assets (honest stats, pseudonymous digest) unblock guardian-consented growth per D14/D18. (3) Laptop demo via PWA by week 10 opens school/partner channels without abandoning consumer-first focus (D17).

Resources. One builder + Claude Code agent fleet (implementation, review, evals). Budget envelope per *PRD.md* §4 unchanged; live-LLM spend stays behind monthly caps (R2).

Success metrics (plan-level). Phase exit criteria below; product KPIs per parent §7 pilot criteria (streak adoption, session frequency, mastery movement).

2. Problem Definition (compressed)

Dimension	Statement
Who	Grade-3 learners (5–9) on shared/borrowed Android phones; caregivers supervising lightly.
What, When, Where	Daily after-school study that must survive dead zones, load-shedding and shared devices; phone-first, home-first (D16).
Why it bites	Prior art punishes poverty: streak loss on a normal Kenyan Tuesday (§3.2), vanity time-metrics, fake progress numbers — all erode child trust exactly when habit formation matters most.
Cost of not solving	Cold-start emptiness plus dishonest gamification means churn before the content pack reaches critical mass (R5).

Full problem framing, market case and decisions D1–D19: see the parent V2 product plan; engineering history: *PRD.md*.

3. Solution Overview & Scope

Evolve the proven offline spine with a habit shell (streaks with grace, mastery-only XP, personal milestones), surface only provably true data, then extend the same backend to caregivers (digest) and laptops (PWA). No leaderboards, no hearts/lives, no raw-activity XP — ever.

4. Timeline — the Twelve Weeks (dated)

Week 1 begins Monday 24 Aug; all dates derive from the 25 Aug 2026 baseline. Overlaps mirror the parent plan's intentional phase overlap.

Week	Focus	Key deliverables	Exit criteria
W1 24 Aug – 30 Aug	Phase 0 closeout — hygiene sweep	Verify OCR lesson pack stays out of public builds; decide provider-abstraction scaffold; pick content-pack growth source; delete orphaned LessonCompleteScreen + route.	Public-safe APK contents confirmed; stray-code debt zero.
W2 31 Aug – 06 Sep	Interests step in onboarding (D7)	Chips-first interest picker at onboarding; server-side mapping to bounded analogy domains; sanitizer tests extended.	New learner completes onboarding with ≥ 3 chips; no free-text path ships.
W3 07 Sep – 13 Sep	Habit polish + regression	Milestone-ceremony animation pass (springs/stagger, low-end-friendly); offline fun-loop E2E regression sweep; Progress/Profile honesty re-audit.	Fun loop usable offline end-to-end (Phase 1 exit criterion).
W4 14 Sep – 20 Sep	L2 evaluation gate	Build/run the ≥ 25 -case adversarial prompt-eval harness mirroring MODEL_EVALUATION.md discipline; verify contract tests + graceful offline fallback.	Eval $\geq$ agreed bar; Phase 1 formally closed; Phase 2 exit criteria met.
W5 21 Sep – 27 Sep	Caregiver linking foundations	Consent-code data model; link/unlink endpoints; pseudonymous-by-default storage checks.	Link/unlink unit-tested; zero PII beyond chosen alias stored.
W6 28 Sep – 04 Oct	Weekly digest job + rendering	Scheduled digest computation over Room-derived truth (topics covered, mastery movement, not-enough-evidence markers); in-app digest screen.	Digest renders honestly for a seeded week of interactions.
W7 05 Oct – 11 Oct	Digest delivery + privacy audit	Email path; consent-gated sending; pseudonymity audit vs D15; Phase 3 exit review.	Link/unlink tested end-to-end; digest contains zero unconsented PII; Phase 3 closed.
W8 12 Oct – 18 Oct	PWA scaffold	Login-by-code; read-only progress + streak views reading the same backend.	Laptop browser shows truthful learner progress.
W9 19 Oct – 25 Oct	PWA practice mode	Quick-check practice against shared backend; offline-tolerant UX.	Practice session completes on laptop over flaky network sim.
W10 26 Oct – 01 Nov	PWA parity + smoke suite	Streak/XP parity rules; automated parity smoke suite; Phase 4 exit review.	Parity suite green; laptop demo possible (Phase 4 closed).
W11 02 Nov – 08 Nov	Hardening fortnight begins	Kenya DPA 2019 checklist walk-through; live-LLM monthly spend cap enforcement verified; public-safe release APK audit.	Compliance checklist signed off internally; caps proven by test.
W12 09 Nov – 15 Nov	Launch readiness	Walk parent \$7 pilot success criteria; demo-day dry run (phone + laptop); freeze scope for pilot.	Go/no-go memo with evidence links; pilot cohort named.

Milestones

ID	When	Definition of done
M1	end 24 Aug – 30 Aug	Hygiene debt zero; public-safe build proven
M2	end 14 Sep – 20 Sep	Phase 1 + Phase 2 exits: unit-tested streak/XP logic, offline fun loop, L2 eval bar met
M3	end 05 Oct – 11 Oct	Caregiver digest live, pseudonymous, consent-gated
M4	end 26 Oct – 01 Nov	PWA parity suite green — laptop demo possible
M5	end 09 Nov – 15 Nov	Launch-ready go/no-go with evidence

5. Functional Requirements (phase-level)

ID	Requirement	Pri.	Notes / when
FR-A	Habit loop: daily activity builds a streak with a one-week freeze token; XP paid only for LESSON_PASSED / REVIEW_STABILIZED / FIRST_TRY_CORRECT_AFTER_STRUGGLE; personal milestones only (no leaderboards)	P0	Landed 25 Aug; REVIEW_STABILIZED emitter deferred until SM-2 scheduler exists
FR-B	Progress/Profile surfaces show exclusively Room-derived truth; '—' means not-enough-evidence	P0	Landed 25 Aug
FR-C	Onboarding captures interests as theme chips; every chip maps server-side to bounded analogy domain	P0	W2
FR-D	/tutor/reexplain passes adversarial eval >= agreed bar; deterministic fallback when provider absent	P0	W4 gate
FR-E	Caregiver link by consent code; weekly digest shows topics covered, mastery movement, honest gaps	P1	W5–7
FR-F	PWA: login-by-code, progress, streak, practice mode reading the production backend	P1	W8–10
FR-G	Multi-day no-connectivity streak awareness	P2	Explicitly deferred (plan §8 amendment, 25 Aug 2026)

6. Non-Functional Requirements

Area	Bar
Offline	Every learner-facing flow in W1–W3 completes with airplane mode on; sync remains opt-in demo-only.
Performance	Ceremony animations spring-based only; smooth on entry-level Androids (no blur/heavy shaders).
Privacy	Pseudonymity until guardian-consented linkage (D15); Kenya DPA 2019 checklist before any public release (R6).
Cost	Live-LLM calls gated behind response caching ≤55-word outputs and monthly caps (R2).
Verification	./gradlew.bat testDebugUnitTest + Django manage.py test stay CI-green every landing (verify-demo.yml).

7. Risks & Mitigations

ID	Risk	Prob.	Impact	Mitigation
R8	Solo-builder bandwidth across overlapping phases	High	Medium	One deliverable-set per week; 20% buffer inside W11; descope animation polish before digest scope

ID	Risk	Prob.	Impact	Mitigation
R2	LLM API cost escalation	Medium	High	Caps enforced by test in W11; open-weight fallback documented
R4	L2 output quality below eval bar at W4	Medium	High	Deterministic fallback keeps feature shippable; eval failures block live-provider enablement only
R5	Cold-start emptiness	Medium	Medium	Habit loop carries value while rule-card pack grows (provenance-gated)
R6	Child-privacy misstep in digest/PWA	Low	Critical	Pseudonymity audit W7; DPA checklist W11; zero-PII digest test
R9	Scope creep from 'Duolingo-grade polish'	High	Low	Polish capped at spring/stagger idioms already shipped; new motion needs explicit approval

8. Explicitly Out of Scope (this 12 weeks)

Leaderboards/child competition (D13); multi-day offline streak awareness (deferred by amendment); hearts/lives punishment mechanics (D12); teacher/classroom channel (D17); L1 BKT model (post-launch phase 5); L3 anything (gated per §6); adult dashboards leading with vanity metrics (D14).

9. Team & Resources

One human builder (product + engineering) with Claude Code as implementation/review fleet: TDD cycles, adversarial diff reviews (the pattern that caught ten defects on 25 Aug including DST streak corruption and an unbuildable spec promise), and eval-harness construction. Infrastructure budget per *PRD.md* §4; no new paid services introduced before M3.

10. Appendix — Source Documents & Decision Log Touchpoints

- HOME_STUDY_COMPANION_V2_PRODUCT_PLAN.md — parent phases §9, gamification rules §8, risks §10, verification §11
- PRD.md — original engineering PRD; its 12-week timeline describes the pre-V2 architecture and is retained as history
- RULE_BASED_TUTOR_MVP.md — content-pack growth order referenced by W1 decision
- MODEL_EVALUATION.md — discipline mirrored by the W4 adversarial eval gate
- Amendment log: plan §8 amended 25 Aug 2026 (streak-grace wording); D7 amended 23 Aug 2026 (chips-first interests)